Reducing Blast Vibration Amplitudes Close-In to Structures of Sub-Standard Construction

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Abstract

Production blasts at one of Copelmi's coal mine operations will be carried out 100 m (330 ft) from a neighborhood located in the town of Butiá, southern Brazil. Mine management is concerned about the possibility of damages and complaints due to blasting operations at close distances to structures of sub-standard construction. Blasting in the area is characterized by the use of very low powder factor resulting in confined shots in which movement is restricted to a point of almost undershooting the rock. Over the past year, tests were conducted mainly in the overburden area to determine the best approach to be implemented when blasts move closer to the town. Mine management wants vibration levels to be reduced lower than the damage limits proposed by Brazilian regulations to ensure a safety margin and mitigate perception by the community hoping to reduce the number of complaints.

Modifications to blast designs included changes in the burden and spacing with the goal of decreasing blast confinement and increasing powder factor, the use of different combinations of delay periods and the introduction of a cast primer replacing emulsion as the initiating charge. Analysis was conducted for every blast design change to evaluate the impacts on vibration levels and blasting operation costs.

This paper presents the challenges faced in order to blast close-in to a residential area and test results for various approaches used to mitigate blast vibrations along with the impacts on the operation cost. Initial results show a reduction in close-in ground vibrations with tighter blast patterns. Smaller burden and spacing resulted in a decrease in blast confinement and an increase of powder factor with an increase in cost of explosives, accessories, and labor.

Introduction

Copelmi Mineração Ltda. has mined in the vicinity of Butiá in Rio Grande do Sul for over forty years and has often faced the challenge of blasting close to residential areas. In the past, regulations and the environmental agencies in Brazil were not so rigid regarding blasting activities. In addition, community members were not so concerned about blasting as they are today. In the last decade the regulatory agencies have restricted vibration and air-pressure (airblast) levels produced by blasting in urban areas, as a result of complaints and lawsuits. Many of these complaints are the result of past practices that are still being applied today by many blasters in the country.

Foreseeing the troubles caused by blasting near neighborhoods, including cessation of operations, mine management started a renewal process in the drilling and blasting practices. The first part of this process was to optimize and standardize drilling and blasting operations (Rosenhaim *et al*, 2012) and in the second part to reduce blasting vibrations for blasting as close as 100 m (330 ft) to structures of sub-standard construction. The structures present in the residential area, addressed here as “structures of sub-standard construction”, are residences built without adherence to construction standards and in many cases there are no “as-built” drawings. Most structures are made of wood, some are built on piers, others with perimeter wall foundations and some with no foundations at all. Others comprise load-bearing clay bricks with mortar. It is this type of construction from which most complaints of cosmetic damages occur.

The Brazilian standard regulation for blasting in urban areas (NBR 9653, 2005) sets the limits for peak particle velocity according to peak frequency as shown in Figure 1. For peak frequency between 4 and 15 Hz, peak particle velocity ranges from 15 mm/s (0.590 in/s) to 20 mm/s (0.787 in/s); for peak frequency from 15 to 40 Hz, peak particle velocity ranges from 20 mm/s (0.787 in/s) to 50 mm/s (1.968 in/s) and above 40 Hz, the peak particle velocity limit is a constant 50 mm/s (1.968 in/s). For peak frequencies below 4 Hz, the criterion is set at a constant 0.6 mm (0.023 in).

As stated by Siskind (2000), ground vibrations and/or airblast well below regulated limits can cause structure motions resulting in wall rattling and structure interior noise. These effects are noticed by homeowners and can cause concern about potential damage. With this in mind, the engineering staff decided that ground motion levels measured at the stand-off distance of 100 m (330 ft) should not exceed 15 mm/s (0.59 in/s) regardless of the peak frequency with the caveat that ground vibrations should be as low as possible.

Figure 2 shows a map with the location of the residential area with respect to the mine project. It was decided that structures closer to the blasting operations would be surveyed for pre-existing damage before the mine operation approached the structures. The yellow line represents the 300 m (985 ft) zone within which all structures should be surveyed as long as the homeowners agree. A total of 150 structures were selected in the area.

Blast Design

The B3 Mine is a multi-seam strip mine utilizing traditional truck and shovel operations to remove the overburden and selectively mine the coal. The overburden comprises a layer of topsoil, red and yellow clay, and siltstone. Four coal seams are mined with interburdens of siltstones. Coal seam thicknesses range from 30 cm to 2.5 m (1 to 8.2 ft) with interburdens ranging from a few centimeters to more than 12 m (40 ft). The overburden is more than 30 m (100 ft). Only a part of the overburden needs to be drilled and blasted because much of the siltstone is soft. Currently, the layers that need blasting are pre-defined and vary in thickness depending on location in the mine site.

Coal seams are blasted along with a portion of siltstone on top to ensure the explosive column is contained within most of the coal seam, resulting in better coal fragmentation. Powder factors used are very low, around 0.155 kg/m^3 (0.261 lbs/yd^3) in the siltstone and 0.280 kg/m^3 (0.472 lbs/yd^3) in the coal. Only sufficient energy is used to fracture the material in place, with little movement, preventing coal dilution.

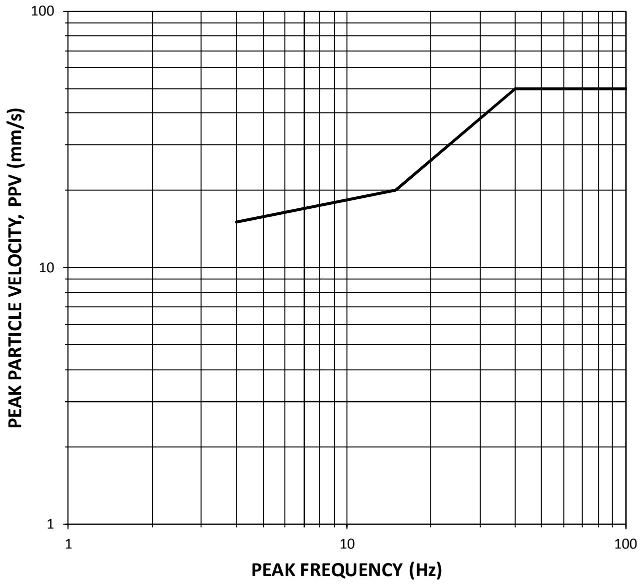

Figure 1: Peak particle velocity versus peak frequency, Brazilian standard regulation for blasting, ABNT NBR 9653:2005

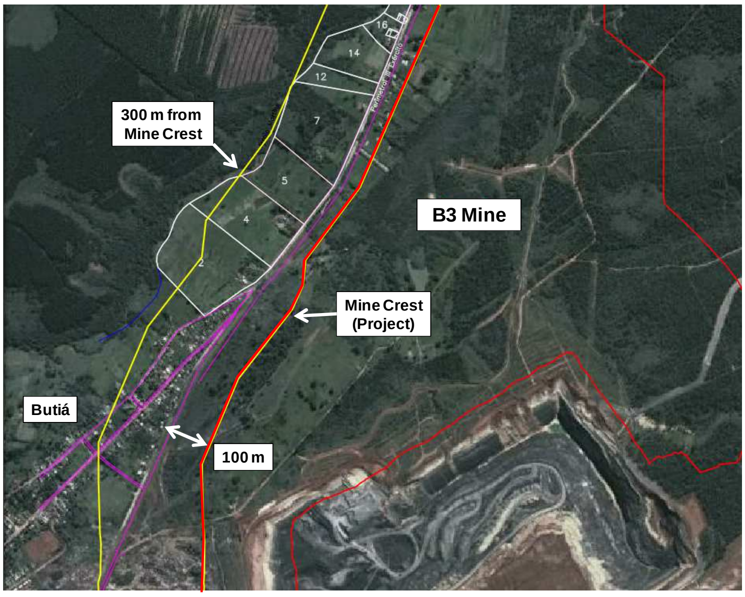

Figure 2: Location map showing the residential area and the B3 Mine

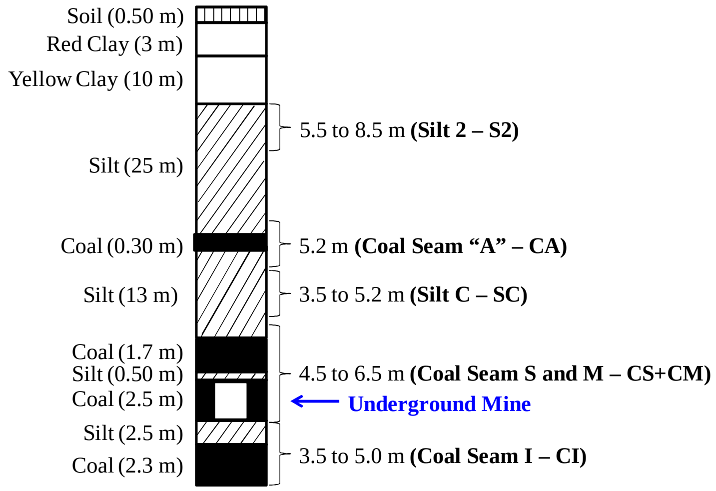

Figure 3: Coal deposit profile with five established blast benches Silt 2, Coal Seam "A", Silt C, Coal Seam S and M and Coal Seam I

Figure 3 shows the profile for the B3 mine coal deposit with five blast benches of which three are considered overburden and two for the major coal seams. It is important to note that one of the major coal seams has been previously underground mined and today the remaining coal pillars are being extracted.

Holes, 76 mm (3") in diameter, are drilled in a staggered pattern and lengths vary according to coal or siltstone thickness, as shown above. Burden and spacing for overburden benches are typically 4 x 5 m (13 x 16.5 ft) and 3 x 4 m (10 x 13 ft) and 3 x 3.5 m (10 x 11.5 ft) for CS+CM and CI benches, respectively.

Explosives used are emulsion cartridges and bulk ANFO. A one-quarter (¼) emulsion cartridge is used as a primer to initiate the ANFO column (Munaretti, 2002). Shock tubes with and without delay timing are used to initiate in-hole primers and surface connections. The low-cost non-electric delay periods accuracies were measured and show a scatter in timing up to 46 % (Rosenhaim, internal study) making the maximum number of holes per delay very difficult to predict. A diagonal initiation sequence is used where a control line is placed parallel to the bench face along the first drilled line and hole-to-hole lines are connected diagonal to the control line. Typically delay periods used in the control line and hole-to-hole connections are each 30 milliseconds (ms), which results in one hole firing at the same time in each line.

Test Bench

Table 1 presents the elevation, relative to sea level, and depth from the surface for each blast bench. The approximate horizontal distances from each blast bench to the nearest structures are estimated, when operations are 100 m (330 ft) from structures. As bench depth increases, the distance of the blasts to the structures also increases. Therefore, ground vibrations from lower bench blasts may not be of great concern based on attenuation from longer horizontal distance. Attenuation of ground motions for coal shots show lower vibration levels manly due to the lower confinement (tighter patterns and higher powder factors). Figure 4 shows a comparative attenuation plot for overburden benches and coal benches for overall data gathered during monitoring.

Table 1: Summary of information for each blast bench

Material	Elevation (m)	Depth (m)	Distance to Strucutres (m)
Surface	50	0	100
S2	32	18	127
CA	15	35	144
SC	8	42	151
CSM	3	47	156
CI	-3	53	162

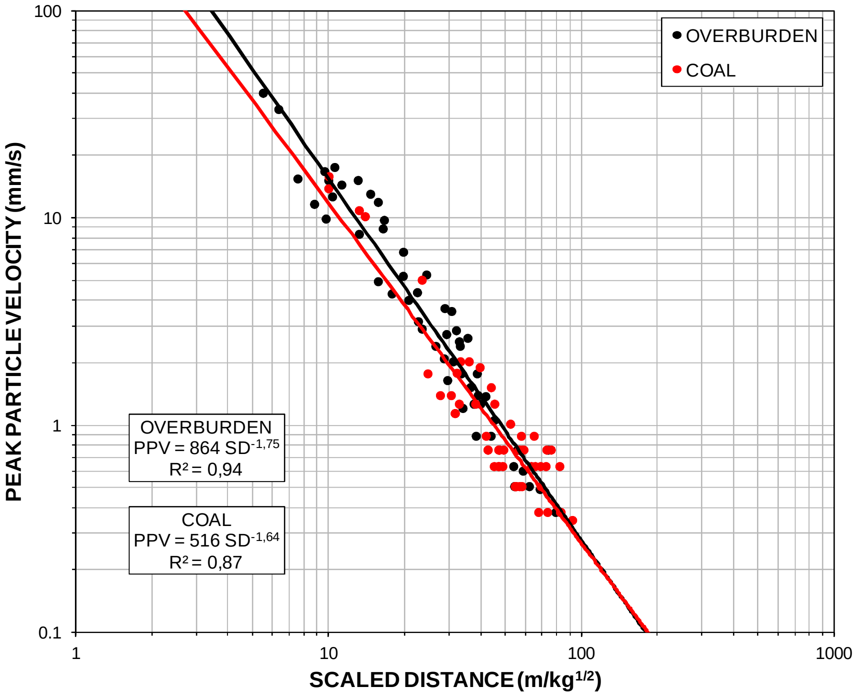

Figure 4: Attenuation plot for overburden and coal blasts, overall data for typical blasts design

The top most overburden blast bench, S2, was selected to conduct the tests presented in this paper. This bench was selected because it is the closest to the surface and ground vibrations are not affected by slope height as there is little attenuation of ground motions with depth. Another reason to apply changes to blast design in only one blast bench is to limit any impacts on total operational cost. Once the optimum design is determined for this bench, it can be reproduced for other benches.

Results

Ten comparative blasts in two series were conducted each with the goal to reduce ground vibration amplitudes. In Series A, changes in the blast design included reducing the number of holes affecting the maximum charge weight per delay, reducing the powder factor by the application of a deck in the explosive column, and reducing the burden and spacing. Series B included replacing emulsion cartridges with cast boosters as primers then varying drill pattern geometries and using more accurate detonators (delay period scatter up to 17% based on industry information).

Series A

Reducing the number of holes per delay (3 blasts)

In many cases, blast patterns are drilled in four up to 15 lines with 10 up to 20 holes per line. The average number of holes per delay for the S2 bench is six with a maximum of 12 while the average charge weight per delay is 100 kg (220 lbs) with a maximum of 200 kg (440 lbs).

To reduce these numbers it was decided to "break" the shot using a long delay period in between a group of lines. In these tests, four lines were tied together and a delay period of either 100 ms or 300 ms was used to separate the next four lines in the pattern. The use of a 300 ms delay period reduced the maximum number of holes per delay to four and the 100 ms delay resulted in six holes per delay. This approach had the lowest impact on cost, as only a few extra delay periods were used and there were no major changes in the blast design.

Decking charges (3 blasts)

The single explosives column of a typical blast design was separated into two decks without changing the top stemming length. A 0.5-m (1.65-ft) length of drill cuttings was placed 3.5 m (11.5 ft) from the blast-hole collar in a 5.5 m (18 ft) hole. This practice reduced the powder factor. However, this increased the blasting cost 1.8% due to the need for one additional cap and primer per hole as well as more time to prepare the blast, because closer attention is required by the blasters when loading the holes.

Changing drill pattern geometry (5 blasts)

It was decided to reduce burden and spacing for the silt overburden bench using the same drill pattern as for the coal shots, 3 x 4 m (10 x 13 ft), because coal shots produced lower ground vibration, as shown in Figure 4. However, this modification increased both the number of blast holes and powder factor thereby increasing the blasting costs by 80%. This approach is considered to be impractical.

The above mentioned tests were compared using ground vibration attenuation plots in Figure 5. The decrease in the number of holes per delay resulted in less confined blasts shown by a lower y-intercept factor, with lower vibrations at the standoff distance (cases 3 and 4). The use of decks resulted in more confined shots with higher close-in vibrations due to a lower powder factor (case 2). The blasts with higher powder factors (case 5), resulting from reduced burden and spacing distances, showed a significant decrease in blast confinement and in ground vibrations at 100 m (330 ft), shown in Table 2, when compared to the typical blast design (case 1).

Table 2: Peak particle velocity at 100 m standoff distances comparing Series A blasts

Case Number	**PPV at 100 m (mm/s)**
Case 1	15.94
Case 2	17.20
Case 3	11.18
Case 4	10.29
Case 5	9.92

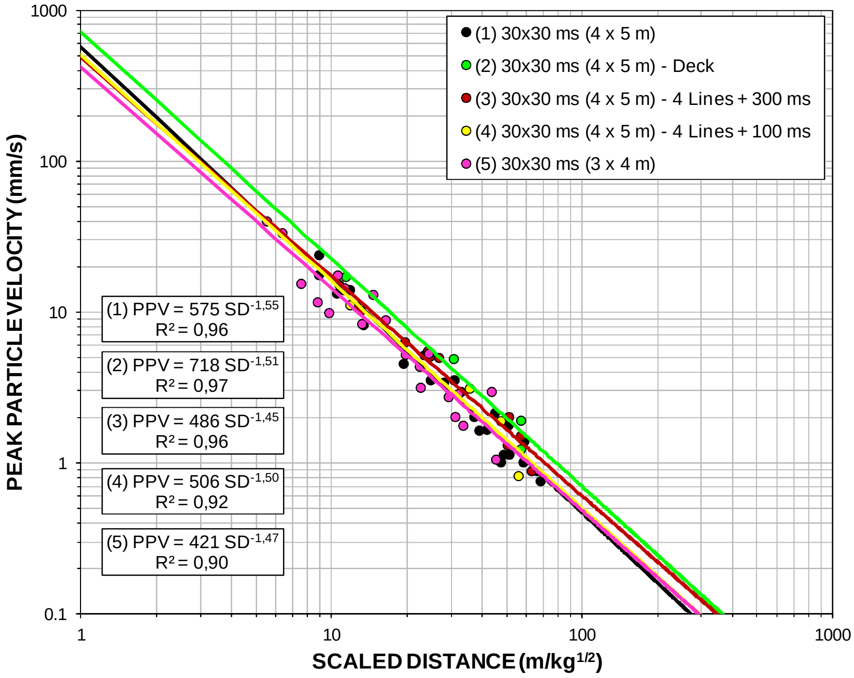

Figure 5: Attenuation plot comparing typical blast to lower number of holes per delay, decking and change in drill pattern geometry

<u>Series B</u>

Series A blasts utilized emulsion as the primer. In Series B the performance of emulsion primers and two sizes of cast boosters were compared using single-hole tests to select the optimum primer. Once selected, certain tests from Series A were repeated to measure the effects of blast design modification on ground vibrations.

Boosters type and size (6 blasts)

Three single-hole tests were conducted to evaluate the performance of cast boosters and emulsion used as primers in a fully-loaded blast hole, 5 m (16.5 ft) in depth, and included a typical one-quarter emulsion cartridge (570 g), 30 g and 150 g pentolite boosters. The blasted material was excavated with a Volvo EC460C excavator. The damaged surface area created by the single-hole blast was measured along with the crater diameter and excavated depth. Table 3 presents the data for each test showing the amount of explosives in each hole, drilled depth, resulting peak particle velocity and distance from the test to the seismograph.

The 150 g cast booster resulted in the deepest crater with a slightly higher crater diameter and the lowest ground vibration levels, given that the seismograph was placed closest to this test. The fragmentation for this blast, from visual inspection measurement was not possible at this time), was smaller than the blast initiated with the emulsion cartridge and the 30 g booster. Therefore, it was decided to use the 150 g booster for subsequent tests and for the analyses presented here it is referenced only as the “booster”.

Table 3: Summary of crater tests design and results

Type of Booster Charge	Hole Depth (m)	Hole Charge (kg)	Crater Diameter (m)	Crater Depth (m)	Area of Surface Damage Befor Excavaton (m^2)	Distance to Seismograph (m)	Scaled Distance ($m/kg^{1/2}$)	PPV (mm/s)
1/4 Emulsion	5.1	14.6	13.0	3.0	26.0	74	19.4	10.4
150 g Booster	5.0	13.8	13.3	3.3	28.6	65	17.5	10.5
30 g Booster	5.2	14.5	13.0	2.6	44.1	75	19.7	10.4

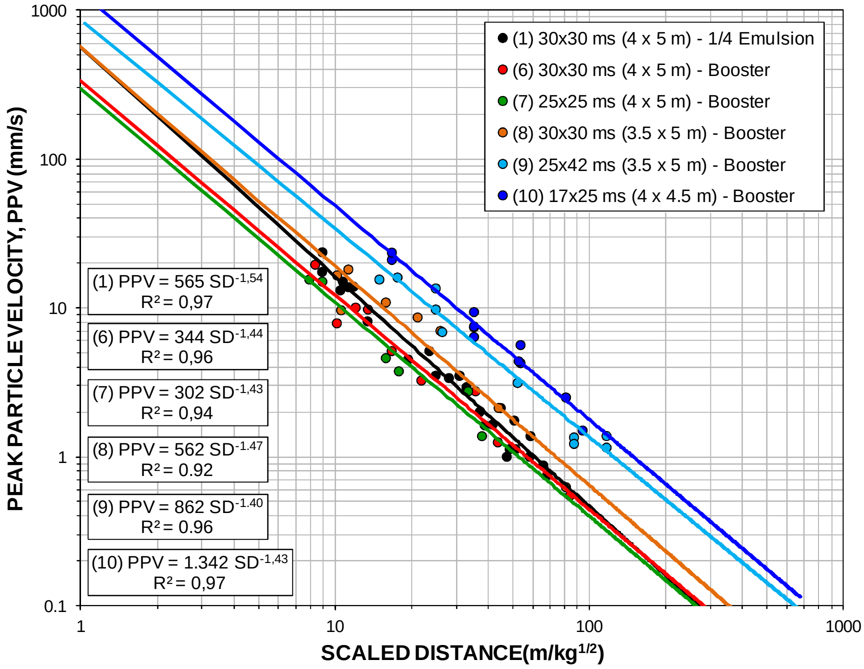

Figure 6: Attenuation plot comparing changes in blast design.

A series of blasts were conducted with the booster and many of the blast designs from Series A. The results of these tests are summarized in Figure 6 comparing confinement (e.g. the y-intercept) and slope of the attenuation plots for blasts of the same design. The introduction of the booster (case 6) increased costs 7.5% which can be considered acceptable because of the reduction in vibration levels as well as the better fragmentation of the material.

Changing drill pattern geometry (4 blasts)

In these tests (case 8) burden was reduced by 0.5 m (1.65 ft) and spacing remained the same from case 1 to reduce confinement. The attenuation curve in Figure 6 shows no change in the y-intercept and a slight decrease in the attenuation slope. This indicates that higher ground motion amplitudes were carried further away from the blast. This design resulted in a 22% increase in cost.

Application of accurate delay periods (3 blasts)

Tests were conducted using more precise delay initiators to measure the effect on ground vibrations. The blast design substituted 30 x 30 ms delays in the case 6 design with 25 x 25 ms delays without changing pattern geometry. This effectively achieved a better prediction of the maximum number of holes per delay. Figure 6 shows that more accurate delay detonators (case 7) resulted in little or no change in the attenuation of ground motion compared with case 6. The cost increase was around 42%. Therefore, the use of precise detonators is not warranted based on the cost increase.

Application of accurate delay periods and changing drill pattern (7 blasts)

A series of tests were conducted using precise delay and combinations of 25 x 42 ms and 17 x 25 ms with modifications to the drill pattern. The burden was first reduced 0.5 m (1.65 ft) from 4 x 5 m (13 x 16.5 ft) to 3.5 to 5 m (11.5 x 16.5 ft). Thereafter, the spacing was reduced from 5 m (16.5 ft) to 4.5 m (14.8 ft) with burden remaining at 4 m (13 ft). These combinations of delay periods resulted in two holes per delay, which is far less than cases 1 and 6. The results, presented in Figure 6, show that both tests produced more confinement (e.g., higher y-intercepts) with higher ground motion amplitudes for all scaled distance values. These tests resulted in a cost increase of 62% for the 25 x 42 ms case 9 and 57% for the 17 x 25 ms case 10.

Comparison of vibrations measured at 100 m (330 ft) standoff distance for nine cases

Figure 7 shows a plot of peak particle velocity versus peak frequency for measurements at 100 m (330 ft) standoff distance from blasts within the Brazilian and USBM safety blasting criteria. The limit set by the mine management of 15 mm/s (0.59 in/s) is also shown and was only fully achieved (100% of data below the line) when using boosters as primers in the original blast design (case 6). For the other tests, ground vibration amplitudes fell below the Brazilian limits due to high frequencies but fell above the mine limit that ignores frequency. With the introduction of boosters as priming charges, vibration levels at the 100 m (330 ft) distance were reduced from an average of 17.5 mm/s (0.69 in/s), for the typical blast, to an average of 9.1 mm/s (0.36 in/s) when using boosters as primer charges.

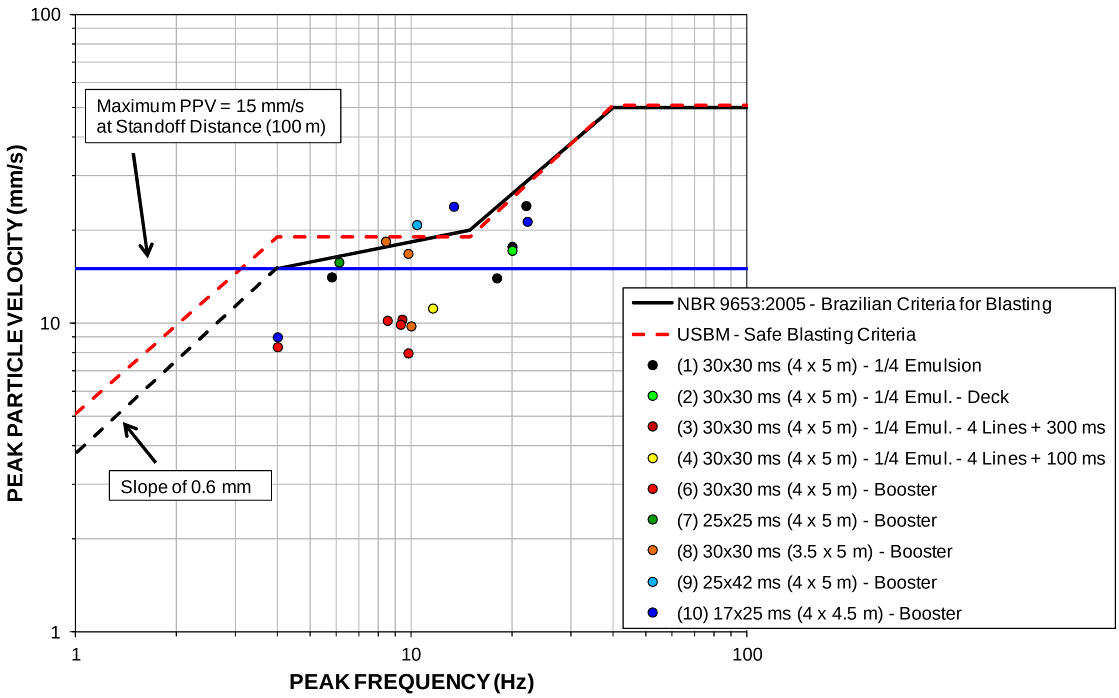

Figure 7: Peak particle velocity versus peak frequency at 100 m (330 ft) standoff distance

Conclusions

These tests show that a reduction in ground vibrations with a minimum impact on operational costs can be achieved with careful evaluation of different blast designs. Sometimes obvious blast design modifications, such as reducing the burden and spacing to decrease blast confinement resulting in lowered blast vibration, can lead to increased operating costs. With a broad analysis of the available resources it is possible to find more suitable and sometimes better alternatives to regulatory constraints when blasting near residential structures. This process serves as a good example of planning for the mitigation of blasting complaints.

It is envisioned that once blasting operation move further away from structures, it will be possible to increase drilling patterns without much effect on vibrations and fragmentation to compensate for the increase in costs of cast boosters.

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